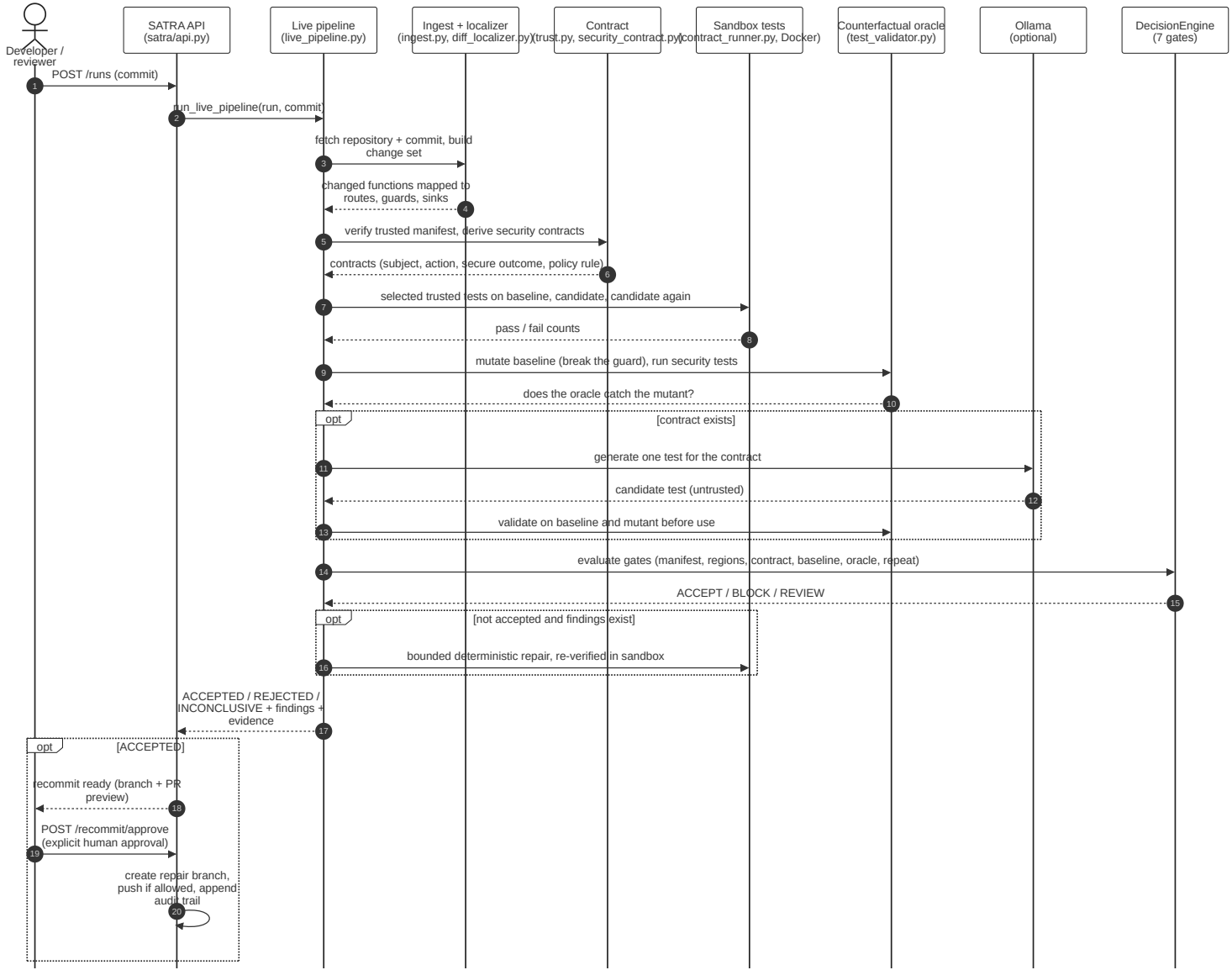

Developer / reviewer
SATRA API (satra/api.py)
Live pipeline (live_pipeline.py)
Ingest + localizer (ingest.py, diff_localizer.py)
Contract (trust.py, security_contract.py)
Sandbox tests (contract_runner.py, Docker)
Counterfactual oracle (test_validator.py)
Ollama (optional)
DecisionEngine (7 gates)
1 POST /runs (commit)
2 run_live_pipeline(run, commit)
3 fetch repository + commit, build change set
4 changed functions mapped to routes, guards, sinks
5 verify trusted manifest, derive security contracts
6 contracts (subject, action, secure outcome, policy rule)
7 selected trusted tests on baseline, candidate, candidate again
8 pass / fail counts
9 mutate baseline (break the guard), run security tests
10 does the oracle catch the mutant?
opt [contract exists]
11 generate one test for the contract
12 candidate test (untrusted)
13 validate on baseline and mutant before use
14 evaluate gates (manifest, regions, contract, baseline, oracle, repeat)
15 ACCEPT / BLOCK / REVIEW
opt [not accepted and findings exist]
16 bounded deterministic repair, re-verified in sandbox
17 ACCEPTED / REJECTED / INCONCLUSIVE + findings + evidence
opt [ACCEPTED]
18 recommit ready (branch + PR preview)
19 POST /recommit/approve (explicit human approval)
20 create repair branch, push if allowed, append audit trail